LEARNING PLAN
NETWORK DESIGN AND MANAGEMENT

UNIT OF COMPETENCE	**Network design & management**
UNIT CODE	0612 451 07A
NAME OF THE TRAINER	JAMES NDIRANGU
COMPETENCE LEVEL	LEVEL 5 MODULE 3
INSTITUTION	THE ABERDARE TECHNICAL COLLEGE
Number of learners	3

SKILL OR JOB TASK

Designing computer network, installing computer network, testing computer network and performing computer network maintenance.

BENCHMARK OR CRITERIA TO BE USED
1. Design computer network *1.1 User needs are collected as per the workplace procedure.* *1.2 Physical network design is developed as per user requirements.* *1.3 Logical network design is developed as per user requirements.*

1.4 Computer network design is mapped out as per user requirements.
2. Install computer network
2.1 Safety measures are observed as per workplace procedure. *2.2 Computer network components are identified as per the network design.* *2.3 Computer network is set up as per the network design.* *2.4 Computer network is configured as per the network design.* 2.5 Computer network is documented as per the network design. 2.6 Computer network comp
3. Test computer network *3.1 Network testing tools and equipment are identified as per the work requirement.* *3.2 Network components are tested as per the workplace procedure.* *3.3 Network testing report is developed as per the work requirement.*
4. Perform computer network maintenance. 4.*1 Computer network maintenance schedule is prepared as per workplace procedure.* *4.2 Computer network is monitored as per maintenance schedule.* *4.3 Computer network is optimized as per the user requirements.*

4.4 Computer network maintenance report is developed as per workplace procedure.

10-WEEK LEARNING PLAN

Week	Session Title	Learning Outcomes (Ordered Bloom's Taxonomy)	Trainer Activities (Active Learning)	Trainee Activities (Active Learning)	Resources & Assessments
1	**Design Computer Network**	1. **Define** "Computer Network Design" (Remembering). 2. **Identify** LAN, WAN, PAN, and MAN types (Remembering).	**Socratic Seminar:** Start by asking trainees for their own definitions of "Design." **Visual Aids:** Show network architecture diagrams.	**Brainstorming:** Groups discuss which topology fits a local business scenario. **Role-Play:** Interview a peer to document "client" needs.	Topo diagrams, Interview forms. **Assessment:** Oral topology quiz.

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		3. **Describe** Star and Mesh topologies (Understanding). 4. **Collect** user requirements (Applying). 5. **Analyze** hardware needs (Analyzing).			
2	**Design Computer Network**	1. **Explain** hierarchical vs. flat design (Understanding).	**Guided Practice:** Demonstrate floor plan design using stencils.	**Project Work:** Develop a physical floor plan including data and access point placement.	Graph paper, Stencils. **Assessment:** Completed floor plan.

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		2. **Draft** a physical network floor plan (Applying). 3. **Distinguish** between greenfield and brownfield sites (Analyzing). 4. **Critique** peer designs for scalability (Evaluating). 5. **Construct** a full design proposal (Creating).	**Case Study:** Present a "brownfield" upgrade scenario for a college lab.	**Peer Review:** Evaluate a partner's design.	

Week	Session Title	Learning Outcomes (Ordered Bloom's Taxonomy)	Trainer Activities (Active Learning)	Trainee Activities (Active Learning)	Resources & Assessments
3	**Install Computer Network**	1. **Define** "Computer Network Installation" (Remembering). 2. **List** essential PPE and safety tools (Remembering). 3. **Discuss** fire/electrical safety (Understanding). 4. **Wear** appropriate PPE correctly (Applying).	**Inquiry-Based Learning:** Ask trainees to identify safety risks in cabling. **Demonstration:** Show proper PPE and tool handling.	**Safety Audit:** Walkthrough the lab to identify and log potential electrical or fire hazards.	PPE, First Aid Kit. **Assessment:** Observation checklist.

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		5. **Evaluate** lab ergonomics (Evaluating).			
4	**Install Computer Network**	1. **Identify** connectors and media (Remembering). 2. **Explain** T568A and T568B standards (Understanding). 3. **Terminate** UTP cables (Applying).	**Live Demo:** Terminating and testing a Cat6 cable. **Jigsaw Activity:** Groups specialize in media types (Fiber, UTP, Coaxial).	**Hands-On Workshop:** Terminate straight-through and crossover cables; peer-test results.	Cat6, RJ45, Crimpers, Testers. **Assessment:** Functional cable.

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		4. **Test** cable continuity (Analyzing). 5. **Judge** if a cable meets industry standards (Evaluating).			
5	**Install Computer Network**	1. **State** differences between IPv4 and IPv6 (Remembering). 2. **Explain** CIDR and subnetting (Understanding).	**Gamification:** Conduct a "Subnetting Speed Challenge" on the board. **Flipped Classroom:** Assign IP basics video before session.	**Problem-Based Learning:** Solve complex subnetting puzzles in small teams.	IP Calculators, Lab PCs. **Assessment:** Subnetting worksheet.

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		3. **Assign** static IP addresses (Applying). 4. **Deconstruct** addresses into subnets (Analyzing). 5. **Defend** choice of IP scheme (Evaluating).			
6	**Install Computer Network**	1. **Recall** CLI commands for switches (Remembering).	**Guided Discovery:** Lead a session on basic router CLI and VLAN setup.	**Lab Simulation:** Configure a multi-switch environment and verify routing via "ping" tests.	Managed Switches, Console cables. **Assessment:** Ping success test.

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		2. **Summarize** VLAN and Trunking functions (Understanding). 3. **Configure** RIP or OSPF protocols (Applying). 4. **Analyze** traffic flow across VLANs (Analyzing). 5. **Optimize** routing paths (Creating).	**Simulation:** Demonstrate inter-VLAN routing errors.		

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7	**Install Computer Network**	1. **List** wireless security protocols (Remembering). 2. **Describe** SSIDs and DHCP (Understanding). 3. **Set up** a wireless access point (Applying). 4. **Compare** privileged account types (Analyzing).	**Debate:** Discuss e-waste regulations vs. practical disposal. **Live Setup:** Configure a wireless AP.	**Case Study:** Configure secure "Staff" vs "Guest" Wi-Fi networks and define privileges.	Wireless APs, E-waste guidelines. **Assessment:** Secure AP setup.

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		5. **Design** an e-waste disposal plan (Creating).			
8	**Test Computer Network**	1. **Define** "Network Testing" (Remembering). 2. **Identify** tools like Wireshark and Ping (Remembering). 3. **Use** traceroute to map paths (Applying).	**Inquiry:** Ask trainees how to verify if a network is "slow" vs "broken." **Flipped Classroom:** Assign Wireshark tutorials.	**Diagnostic Lab:** Use Ping and Traceroute to identify where a simulated network break occurs.	Wireshark, Signal tester. **Assessment:** Network test report.

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		4. **Troubleshoot** packet drops (Analyzing). 5. **Generate** a network test report (Creating).			
9	**Perform Maintenance**	1. **Define** "Network Maintenance" (Remembering). 2. **List** monitoring tools like NSLookup (Remembering).	**Inquiry-Based Learning:** Introduce "pre-set" hardware faults for students to find. **Modeling:** Show how to draft a schedule.	**Jigsaw Activity:** Students specialize in one monitoring tool and teach their peers.	NSLookup, Templates. **Assessment:** Resolution of faults.

Week	Session Title	Learning Outcomes (Ordered Bloom's Taxonomy)	Trainer Activities (Active Learning)	Trainee Activities (Active Learning)	Resources & Assessments
		3. **Prepare** a maintenance schedule (Applying). 4. **Diagnose** DNS issues (Analyzing). 5. **Propose** QoS optimization (Creating).			
10	**Perform Maintenance**	1. **Recall** report components (Remembering).	**Project Review:** Provide feedback on lab logs and reports.	**Final Project:** Execute the full maintenance schedule on the lab and compile the final report.	Lab logbooks, QoS guides. **Assessment:** Final Portfolio Review.

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		2. **Explain** the importance of reports (Understanding). 3. **Implement** QoS settings (Applying). 4. **Analyze** stability after maintenance (Analyzing). 5. **Produce** a final maintenance report (Creating).	**Simulation:** Demonstrate network lag vs. QoS.		